

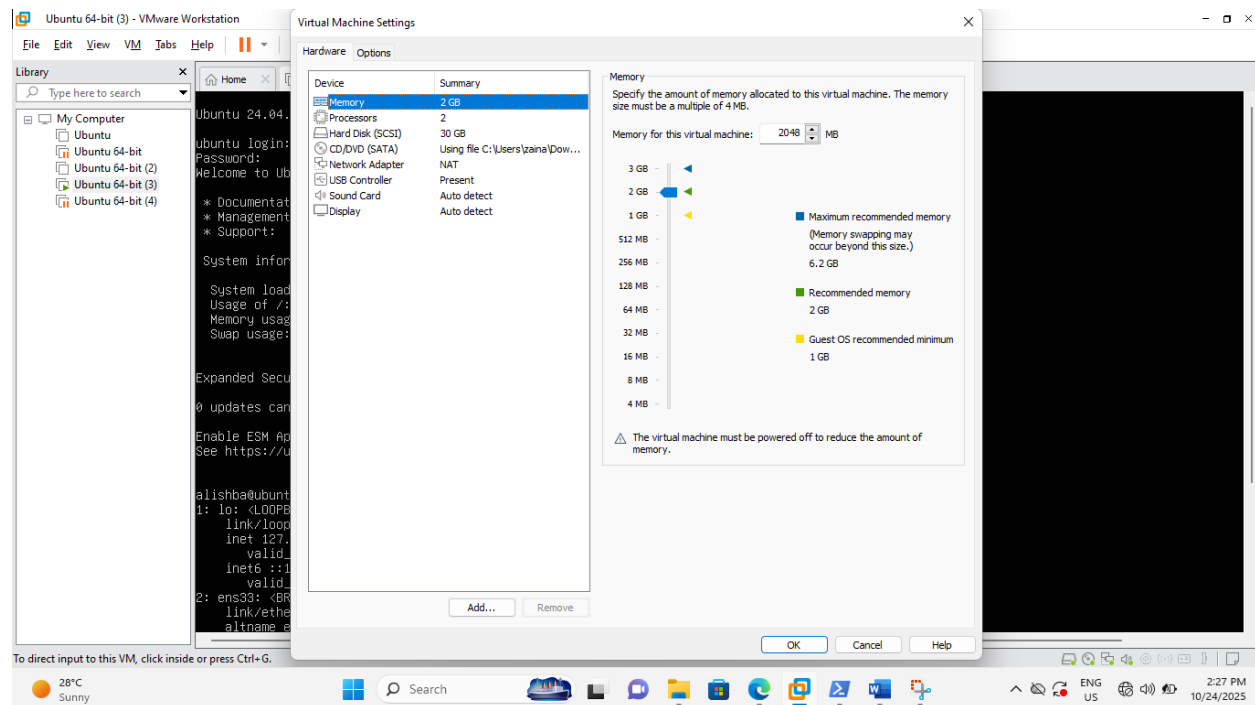
Name: ALISHBA ZAINAB

ROLL NO:23-BSE-006

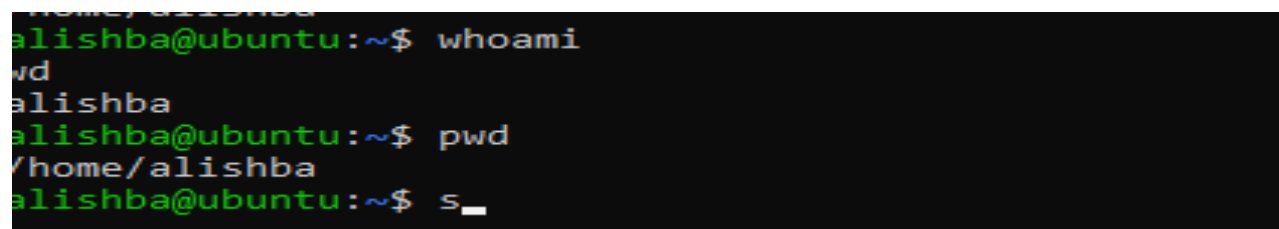
SUBMITTED TO: SIR SHOAIB

Lab 4

Task 1



Task 2



Task 3

Filesystem Exploration

```
alishba@ubuntu: ~  
drwxr-xr-x 3 root root 4096 Oct 3 03:40 systemd-private-85856e69381841b9b8eb35e35a4ac865-systemd-logind.service-1s1eq4  
drwxr-xr-x 3 root root 4096 Oct 24 07:55 systemd-private-85856e69381841b9b8eb35e35a4ac865-systemd-resolved.service-wl8Fbf  
drwxr-xr-x 3 root root 4096 Oct 24 07:55 systemd-private-85856e69381841b9b8eb35e35a4ac865-systemd-timesyncd.service-KBW1DP  
drwxr-xr-x 2 root root 4096 Oct 3 03:40 vmware-root_680-2689143817  
drwxr-xr-x 2 root root 4096 Oct 24 07:55 vmware-root_8441-1680806718  
drwxr-xr-x 2 root root 4096 Oct 24 07:55 vmware-root_8550-995146041  
drwxrwxrwt 2 root root 4096 Oct 24 09:16 .X11-unix  
drwxrwxrwt 2 root root 4096 Oct 3 03:40 .X11-unix  
alishba@ubuntu:~$ ls -la ~  
total 136  
drwxr-xr-x 20 alishba alishba 4096 Oct 24 09:09 .  
drwxr-xr-x 3 root root 4096 Sep 28 15:01 ..  
-rw-rw-r-- 1 alishba alishba 463 Oct 24 08:03 apt_update_vs_upgrade.md  
-rw-rw-r-- 1 alishba alishba 22 Sep 28 15:20 .bash_history  
-rw-rw-r-- 1 alishba alishba 220 Mar 31 2024 .bash_logout  
-rw-rw-r-- 1 alishba alishba 3771 Mar 31 2024 .bashrc  
drwxr-xr-x 8 alishba alishba 4096 Oct 24 09:09 .cache  
drwxr-xr-x 8 alishba alishba 4096 Oct 24 09:09 .config  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Desktop  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Documents  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Downloads  
drwxr-xr-x 3 alishba alishba 4096 Oct 24 08:56 .gnupg  
-rw-rw-r-- 1 alishba alishba 0 Oct 24 08:56 .ICEauthority  
drwxrwxr-x 3 alishba alishba 4096 Oct 24 08:02 .local  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Music  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 .pcsc10  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Pictures  
drwxr-xr-x 3 alishba alishba 4096 Oct 24 09:09 .pki  
-rw-rw-r-- 1 alishba alishba 807 Mar 31 2024 .profile  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Public  
drwxr-xr-x 3 alishba alishba 4096 Oct 24 08:13 snap  
drwxr-xr-x 2 alishba alishba 4096 Sep 28 15:01 .ssh  
-rw-rw-r-- 1 alishba alishba 0 Sep 28 15:20 .sudo_as_admin_successful  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Templates  
drwxrwxr-t 2 alishba alishba 4096 Oct 24 08:56 .thinclient_drives  
drwxr-xr-x 2 alishba alishba 4096 Oct 24 08:56 Videos  
drwxrwxr-x 4 alishba alishba 4096 Oct 24 09:09 .vscode  
-rw-rw-r-- 1 alishba alishba 52 Oct 24 08:56 .Xauthority  
-rw-rw-r-- 1 alishba alishba 21135 Oct 24 09:16 .xorgxrdp-10.log  
-rw-rw-r-- 1 alishba alishba 14 Oct 24 08:42 .xsession  
-rw-rw-r-- 1 alishba alishba 3689 Oct 24 09:16 .xsession-errors  
alishba@ubuntu:~$
```

```
alishba@ubuntu:~$ nano ~/answers.md  
alishba@ubuntu:~$ cat ~/answers.md  
/bin holds essential binaries used by all users (like ls, cp).  
/usr/bin holds most user-level programs installed by the system package manager.  
/usr/local/bin stores custom user-installed software or scripts not managed by the system.  
alishba@ubuntu:~$
```

Task 4

CLI Navigation and File Operations

```
alishba@ubuntu:~$ mkdir -p ~/lab4/workspace/python_project  
alishba@ubuntu:~$ cd ~/lab4/workspace/python_project  
alishba@ubuntu:~/lab4/workspace/python_project$ pwd  
/home/alishba/lab4/workspace/python_project  
alishba@ubuntu:~/lab4/workspace/python_project$
```

```
alishba@ubuntu: ~/lab4/workspace/python_project
GNU nano 7.2
README.md *
hi my name is alishba lab4_
```

Wrote 1 line

^G Help ^O Write Out ^M Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M-A Set Mark ^M-I To Bracket ^M-Q Previous
^X Exit ^R Read File ^N Replace ^J Paste ^_ Justify ^_ Go To Line ^M-E Redo ^M-G Copy ^M-W Where Was ^M-W Next

28°C Sunny Search 2:47 PM 10/24/2025

File 2

```
alishba@ubuntu: ~/lab4/workspace/python_project
GNU nano 7.2
main.py
print("hello lab4")
```

Read 1 line

^G Help ^O Write Out ^M Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M-A Set Mark ^M-I To Bracket ^M-Q Previous
^X Exit ^R Read File ^N Replace ^J Paste ^_ Justify ^_ Go To Line ^M-E Redo ^M-G Copy ^M-W Where Was ^M-W Next

28°C Sunny Search 2:56 PM 10/24/2025

File 3

```
alishba@ubuntu: ~/lab4/workspace/python_project
GNU nano 7.2
hi my name is alishba lab4
README.md
```

Help, Write Out, Where Is, Cut, Execute, Read 1 line, Location, Undo, Set Mark, To Bracket, Previous, Exit, Read File, Replace, Paste, Justify, Go To Line, Redo, Copy, Where Was, Next

28°C Sunny 2:58 PM 10/24/2025

Ls mkdir and list of other functions

```
alishba@ubuntu: ~/lab4/workspace/python_project
*** System restart required ***
Last login: Fri Oct 24 06:53:06 2025 from 192.168.145.1
alishba@ubuntu:~$ nano main.py
alishba@ubuntu:~$ alishba@ubuntu:~$ cd ~/lab4/workspace/python_project$ pwd
-bash: cd: too many arguments
alishba@ubuntu:~$ cd ~/lab4/workspace/python_project
alishba@ubuntu:~/lab4/workspace/python_project$ nano main.py
alishba@ubuntu:~/lab4/workspace/python_project$ nano .env
alishba@ubuntu:~/lab4/workspace/python_project$ alishba@ubuntu:~/lab4/workspace/python_project$ nano README.md
alishba@ubuntu:~/lab4/workspace/python_project$ ls -la
DME.md README.copy.md
mv README.copy.md README.dev.md
rm README.dev.md
mkdir total 20
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:00 .
drwxrwxr-x 3 alishba alishba 4096 Oct 24 09:43 ..
-rw-rw-r-- 1 alishba alishba 24 Oct 24 09:58 .env
-rw-rw-r-- 1 alishba alishba 21 Oct 24 09:57 main.py
-rw-rw-r-- 1 alishba alishba 27 Oct 24 10:00 README.md
alishba@ubuntu:~/lab4/workspace/python_project$ cp README.md README.copy.md
alishba@ubuntu:~/lab4/workspace/python_project$ cp README.md README.copy.md
alishba@ubuntu:~/lab4/workspace/python_project$ mv README.copy.md README.dev.md
alishba@ubuntu:~/lab4/workspace/python_project$ rm README.dev.md
alishba@ubuntu:~/lab4/workspace/python_project$ mkdir -p ~/lab4/workspace/java_app
alishba@ubuntu:~/lab4/workspace/python_project$ cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
alishba@ubuntu:~/lab4/workspace/python_project$ ls -la ~/lab4/workspace
total 20
drwxrwxr-x 5 alishba alishba 4096 Oct 24 10:00 .
drwxrwxr-x 3 alishba alishba 4096 Oct 24 09:43 ..
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:00 java_app
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:00 java_app_copy
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:00 python_project
alishba@ubuntu:~/lab4/workspace/python_project$ history
1 ip addr
2 sudo poweroff
3 nano main.py
4 cd ~/lab4/workspace/python_project$ pwd
5 cd ~/lab4/workspace/python_project
6 nano main.py
7 nano .env
8 nano README.md
```

28°C Sunny 3:02 PM 10/24/2025

Tab completion

```
16 history
shba@ubuntu:~/lab4/workspace/python_project$ cd ~/lab4/workspace/
```

Task 5

```
alishba@ubuntu: ~/lab4/workspace/python_project
alishba 29361 0.0 0.2 9556 4736 ? S 08:56 0:00 /usr/bin/dbus-daemon --config-file=/usr/share/defaults/at-spi2/accessibility.conf --nofork --print-ad
alishba 29381 0.0 0.6 690592 12916 ? Ssl 08:56 0:00 /usr/libexec/xdg-desktop-portal
alishba 29394 0.0 0.3 756188 7040 ? Ssl 08:56 0:00 /usr/libexec/xdg-document-portal
alishba 29401 0.0 0.2 386708 5760 ? Ssl 08:56 0:00 /usr/libexec/xdg-permission-store
root 29415 0.0 0.0 2704 1920 ? Ss 08:56 0:00 fusermount3 -o rw,nosuid,nodev,fsname=portal,auto_unmount,subtype=portal -- /run/user/1000/doc
alishba 29426 0.0 0.2 230108 4992 ? Ssl 08:56 0:00 /usr/libexec/dconf-service
rtkit 29444 0.0 0.1 22940 3328 ? SNI 08:56 0:00 /usr/libexec/rtkit-daemon
colord 29492 0.0 0.6 317468 12412 ? Ssl 08:56 0:00 /usr/libexec/colord
alishba 29510 0.0 1.7 55252 34432 ? S 08:56 0:00 /usr/bin/python3 /usr/share/system-config-printer/applet.py
alishba 29569 0.0 0.4 460296 9728 ? Ssl 08:56 0:00 /usr/libexec/gvfs-udisks2-volume-monitor
alishba 29595 0.0 0.3 307172 6144 ? Ssl 08:56 0:00 /usr/libexec/gvfs-goa-volume-monitor
alishba 29608 0.0 0.3 386776 7168 ? Ssl 08:56 0:00 /usr/libexec/gvfs-afc-volume-monitor
alishba 29617 0.0 0.3 387192 6272 ? Ssl 08:56 0:00 /usr/libexec/gvfs-mtp-volume-monitor
alishba 29622 0.0 0.3 308160 6272 ? Ssl 08:56 0:00 /usr/libexec/gvfs-gphoto2-volume-monitor
alishba 29651 0.0 1.6 1270152 33024 ? Ssl 08:56 0:00 /usr/libexec/evolution-source-registry
alishba 29668 0.0 0.4 532976 8576 ? Sl 08:56 0:00 /usr/libexec/gvfsd-trash --spawner :1.5 /org/gtk/gvfs/exec_spaw/0
alishba 29679 0.0 0.3 233704 6144 ? Ssl 08:56 0:00 /usr/libexec/gvfsd-metadata
alishba 29681 0.0 0.9 1150272 10200 ? Ssl 08:56 0:00 /usr/libexec/evolution-calendar-factory
alishba 29690 0.0 1.2 896548 24064 ? Ssl 08:56 0:01 /usr/libexec/evolution-addressbook-factory
root 29938 0.0 0.0 0 0 ? I< 09:09 0:00 [kworker/u260:3-ttm]
root 30356 0.0 0.0 0 0 ? I< 09:15 0:00 [kworker/1:1H-kblockd]
root 30360 0.0 0.0 0 0 ? I 09:15 0:00 [kworker/u257:3-events_power_efficient]
root 30389 0.0 0.0 0 0 ? I< 09:16 0:00 [kworker/u260:1-ttm]
root 30427 0.0 0.0 0 0 ? I< 09:16 0:00 [kworker/u261:8]
root 30428 0.0 0.0 0 0 ? I< 09:16 0:00 [kworker/u261:9]
root 30490 0.0 0.0 0 0 ? I 09:25 0:01 [kworker/1:0-events]
root 30515 0.0 2.1 477960 41672 ? Ssl 09:29 0:01 /usr/libexec/fwupd/fwupd
root 30550 0.3 0.0 0 0 ? I 09:40 0:04 [kworker/0:2-events]
root 30557 0.0 0.0 0 0 ? I< 09:43 0:00 [kworker/0:1H]
root 30578 0.0 0.0 0 0 ? I< 09:43 0:00 [kworker/u257:1-events_unbound]
root 30709 0.0 0.0 0 0 ? I< 09:50 0:00 [kworker/1:0H]
root 30727 0.0 0.0 0 0 ? I 09:51 0:00 [kworker/0:1-rcu_par_gp]
root 30744 0.0 0.0 0 0 ? I 09:52 0:00 [kworker/u258:0-events_unbound]
root 30770 0.0 0.4 12020 8064 ? Ss 09:53 0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root 30772 0.0 0.5 14992 10624 ? Ss 09:53 0:00 sshd: alishba [priv]
alishba 30858 0.0 0.3 14992 6984 ? S 09:53 0:00 sshd: alishba@pts/1
alishba 30861 0.0 0.3 10124 6400 pts/1 Ss 09:53 0:00 -bash
alishba 30902 0.0 0.4 313880 9600 ? Sl 09:56 0:00 /usr/bin/gnome-keyring-daemon --start --foreground --components=secrets
root 30910 0.0 0.0 0 0 ? I 09:59 0:00 [kworker/u257:0-events_power_efficient]
root 30913 0.0 0.0 0 0 ? I 10:00 0:00 [kworker/1:1-cgroup_destroy]
root 30923 0.0 0.0 0 0 ? I 10:02 0:00 [kworker/u257:2-events_unbound]
alishba 30936 150 0.2 11012 4480 pts/1 R+ 10:06 0:00 ps aux
alishba@ubuntu:~/lab4/workspace/python_project$
```

Task 6

```
alishba@ubuntu: ~/lab4/workspace/python_project
root 30557 0.0 0.0 0 0 ? I< 09:43 0:00 [kworker/0:1H]
root 30578 0.0 0.0 0 0 ? I< 09:43 0:00 [kworker/u257:1-events_unbound]
root 30709 0.0 0.0 0 0 ? I< 09:50 0:00 [kworker/1:0H]
root 30727 0.0 0.0 0 0 ? I 09:51 0:00 [kworker/0:1-rcu_par_gp]
root 30744 0.0 0.0 0 0 ? I 09:52 0:00 [kworker/u258:0-events_unbound]
root 30770 0.0 0.4 12020 8064 ? Ss 09:53 0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root 30772 0.0 0.5 14992 10624 ? Ss 09:53 0:00 sshd: alishba [priv]
alishba 30858 0.0 0.3 14992 6984 ? S 09:53 0:00 sshd: alishba@pts/1
alishba 30861 0.0 0.3 10124 6400 pts/1 Ss 09:53 0:00 -bash
alishba 30902 0.0 0.4 313880 9600 ? Sl 09:56 0:00 /usr/bin/gnome-keyring-daemon --start --foreground --components=secrets
root 30910 0.0 0.0 0 0 ? I 09:59 0:00 [kworker/u257:0-events_power_efficient]
root 30913 0.0 0.0 0 0 ? I 10:00 0:00 [kworker/1:1-cgroup_destroy]
root 30923 0.0 0.0 0 0 ? I 10:02 0:00 [kworker/u257:2-events_unbound]
alishba 30936 150 0.2 11012 4480 pts/1 R+ 10:06 0:00 ps aux
alishba@ubuntu:~/lab4/workspace/python_project$ sudo adduser lab4user
tent passwd lab4user
su - lab4user
sudo whoami # expect permission denied
exit
[sudo] deluser --remove-home lab4user
[sudo] password for alishba:
Sorry, try again.
[sudo] password for alishba:
info: Adding user 'lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'lab4user' (1001) ...
info: Adding new user 'lab4user' (1001) with group 'lab4user' (1001) ...
info: Creating home directory '/home/lab4user' ...
info: Copying files from '/etc/skel' ...
New passwords
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
info: Adding new user 'lab4user' to supplemental / extra groups 'users' ...
info: Adding user 'lab4user' to group 'users' ...
alishba@ubuntu:~/lab4/workspace/python_project$
```

Task 7

Bonus task

```
alishba@ubuntu:~/lab4/workspace/python_project$ alishba@ubuntu:~/lab4/workspace/python_project$ chmod +x ~/lab4/work
~/lab4/workspace/run-demo.sh
sudo ~/lab4/workalishba@ubuntu:~/lab4/workspace/python_project$ ~/lab4/workspace/run-demo.sh
pace/run-demo.sh
Lab 4 demo: current user is alishba
Current time: Fri 24 Oct 10:12:39 UTC 2025
10:12:39 up 3:31, 4 users, load average: 0.00, 0.00, 0.00
      total    used    free   shared  buff/cache   available
Mem:    1.9Gi    492Mi    855Mi    612Ki    755Mi    1.4Gi
Swap:    2.0Gi     68Mi    1.9Gi
alishba@ubuntu:~/lab4/workspace/python_project$ sudo ~/lab4/workspace/run-demo.sh
Lab 4 demo: current user is root
Current time: Fri 24 Oct 10:12:39 UTC 2025
10:12:39 up 3:31, 4 users, load average: 0.00, 0.00, 0.00
      total    used    free   shared  buff/cache   available
Mem:    1.9Gi    493Mi    854Mi    612Ki    755Mi    1.4Gi
Swap:    2.0Gi     68Mi    1.9Gi
alishba@ubuntu:~/lab4/workspace/python_project$ _
```

Exam evaluation questions

Question 1

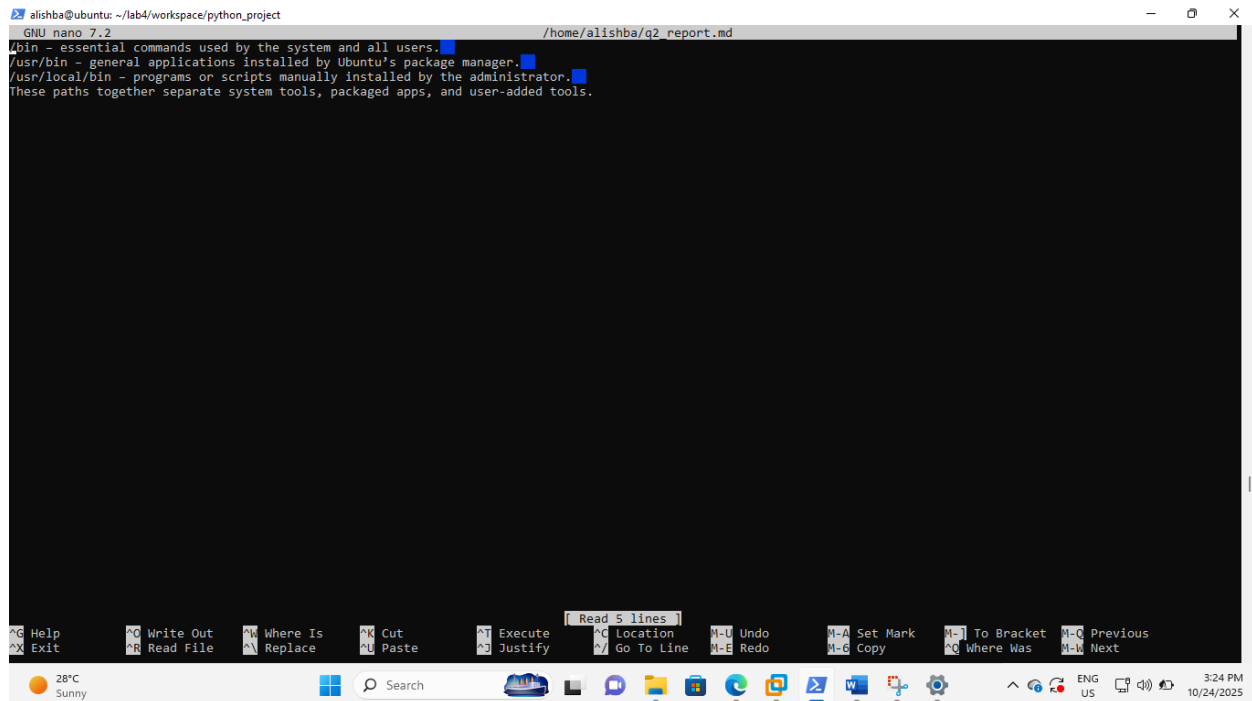
```
alishba
alishba@ubuntu:~/lab4/workspace/python_project$ pwd
/home/alishba/lab4/workspace/python_project
alishba@ubuntu:~/lab4/workspace/python_project$ .
/home/alishba/lab4/workspace/python_project
alishba@ubuntu:~/lab4/workspace/python_project$ hostname
ubuntu
alishba@ubuntu:~/lab4/workspace/python_project$ _
```

Question 2

```
alishba@ubuntu: ~/lab4/workspace/python_project
Lab 4 demo: current user is root
Current time: Fri 24 Oct 10:12:39 UTC 2025
10:12:39 up 3:31, 4 users, load average: 0.00, 0.00, 0.00
Mem:      1.9Gi 493Mi 854Mi 612Ki 755Mi 1.4Gi
Swap:     2.0Gi 68Mi 1.9Gi
alishba@ubuntu:~/lab4/workspace/python_project$ whoami
alishba
alishba@ubuntu:~/lab4/workspace/python_project$ pwd
/home/alishba/lab4/workspace/python_project
alishba@ubuntu:~/lab4/workspace/python_project$ hostname
ubuntu
alishba@ubuntu:~/lab4/workspace/python_project$ ls -la /
total 2097252
drwxr-xr-x 23 root root 4096 Sep 28 14:39 .
drwxr-xr-x 23 root root 4096 Sep 28 14:39 ..
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged
drwxr-xr-x 3 root root 4096 Oct 24 07:55 boot
dr-xr-xr-x 2 root root 4096 Aug 5 23:53 cdrom
drwxr-xr-x 19 root root 4096 Oct 3 03:40 dev
drwxr-xr-x 141 root root 12288 Oct 24 10:10 etc
drwxr-xr-x 4 root root 4096 Oct 24 10:09 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64
drwxr-xr-x 2 root root 4096 Feb 26 2024 lib.usr-is-merged
drwx----- 2 root root 16384 Sep 28 14:37 lost+found
drwxr-xr-x 2 root root 4096 Aug 5 16:54 media
drwxr-xr-x 2 root root 4096 Aug 5 16:54 mnt
drwxr-xr-x 2 root root 4096 Aug 5 16:54 opt
dr-xr-xr-x 312 root root 0 Oct 3 03:40 proc
drwx----- 4 root root 4096 Oct 24 08:40 root
drwxr-xr-x 37 root root 1120 Oct 24 09:53 run
lrwxrwxrwx 1 root root 8 Apr 22 2024/sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 Dec 11 2024/sbin.usr-is-merged
drwxr-xr-x 6 root root 4096 Oct 24 08:09 snap
drwxr-xr-x 2 root root 4096 Aug 5 16:54 srv
-rw----- 1 root root 2147483648 Sep 28 14:39 swap.img
dr-xr-xr-x 13 root root 0 Oct 24 09:36 sys
drwxrwxrwt 20 root root 4096 Oct 24 09:29 tmp
drwxr-xr-x 12 root root 4096 Aug 5 16:54 usr
drwxr-xr-x 14 root root 4096 Oct 24 08:32 var
alishba@ubuntu:~/lab4/workspace/python_project$
```

```
alishba@ubuntu: ~/lab4/workspace/python_project
drwxr-xr-x 23 root root 4096 Sep 28 14:39 .
drwxr-xr-x 23 root root 4096 Sep 28 14:39 ..
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged
drwxr-xr-x 3 root root 4096 Oct 24 07:55 boot
dr-xr-xr-x 2 root root 4096 Aug 5 23:53 cdrom
drwxr-xr-x 19 root root 4096 Oct 3 03:40 dev
drwxr-xr-x 141 root root 12288 Oct 24 10:10 etc
drwxr-xr-x 4 root root 4096 Oct 24 10:09 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64
drwxr-xr-x 2 root root 4096 Feb 26 2024 lib.usr-is-merged
drwx----- 2 root root 16384 Sep 28 14:37 lost+found
drwxr-xr-x 2 root root 4096 Aug 5 16:54 media
drwxr-xr-x 2 root root 4096 Aug 5 16:54 mnt
drwxr-xr-x 2 root root 4096 Aug 5 16:54 opt
dr-xr-xr-x 312 root root 0 Oct 3 03:40 proc
drwx----- 4 root root 4096 Oct 24 08:40 root
drwxr-xr-x 37 root root 1120 Oct 24 09:53 run
lrwxrwxrwx 1 root root 8 Apr 22 2024/sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 Dec 11 2024/sbin.usr-is-merged
drwxr-xr-x 6 root root 4096 Oct 24 08:09 snap
drwxr-xr-x 2 root root 4096 Aug 5 16:54 srv
-rw----- 1 root root 2147483648 Sep 28 14:39 swap.img
dr-xr-xr-x 13 root root 0 Oct 24 09:36 sys
drwxrwxrwt 20 root root 4096 Oct 24 09:29 tmp
drwxr-xr-x 12 root root 4096 Aug 5 16:54 usr
drwxr-xr-x 14 root root 4096 Oct 24 08:32 var
alishba@ubuntu:~/lab4/workspace/python_project$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
alishba@ubuntu:~/lab4/workspace/python_project$
```

```
alishba@ubuntu: ~/lab4/workspace/python_project
GNU nano 7.2 /home/alishba/q2_report.md
/bin - essential commands used by the system and all users.
/usr/bin - general applications installed by Ubuntu's package manager.
/usr/local/bin - programs or scripts manually installed by the administrator.
These paths together separate system tools, packaged apps, and user-added tools.
```



Question 3

```
alishba@ubuntu: ~/analysis/workspace$ cp -r ~/analysis/workspace ~/analysis/workspace_backup
alishba@ubuntu: ~/analysis/workspace$ ls -la ~/analysis
total 16
drwxrwxr-x 4 alishba alishba 4096 Oct 24 10:27 .
drwxr-x--- 22 alishba alishba 4096 Oct 24 10:26 ..
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:26 workspace
drwxrwxr-x 2 alishba alishba 4096 Oct 24 10:27 workspace_backup
alishba@ubuntu: ~/analysis/workspace$
```

```
alishba@ubuntu: ~/analysis/workspace
1 ip addr
2 sudo poweroff
3 nano main.py
4 cd ~/lab4/workspace/python_project$ pwd
5 cd ~/lab4/workspace/python_project
6 nano main.py
7 nano .env
8 nano README.md
9 ls -la
10 cp README.md README.copy.md
11 mv README.copy.md README.dev.md
12 rm README.dev.md
13 mkdir -p ~/lab4/workspace/java_app
14 cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
15 ls -la ~/lab4/workspace
16 history
17 cd ~/lab4/workspace/uname -a
18 cat /proc/cpuinfo
19 free -h
20 df -h
21 cat /etc/os-release
22 ps aux
23 sudo adduser lab4user
24 nano ~/lab4/workspace/run-demo.sh
25 chmod +x ~/lab4/workspace/run-demo.sh
26 ~/lab4/workspace/run-demo.sh
27 sudo ~/lab4/workspace/run-demo.sh
28 whoami
29 pwd
30 hostname
31 ls -la /
32 cat /etc/os-release
33 nano ~/q2/report.md
34 mkdir -p ~/analysis/workspace
35 cd ~/analysis/workspace
36 echo "report data" > report.txt
37 echo "summary text" > notes.txt
38 echo "hidden evidence" > .secret.txt
39 ls -la
40 cp -r ~/analysis/workspace ~/analysis/workspace_backup
41 ls -la ~/analysis
42 history
alishba@ubuntu:~/analysis/workspace$
```

28°C Sunny 3:28 PM 10/24/2025

Question 4

```
alishba@ubuntu:~/analysis/workspace$ uname -a
Linux ubuntu 6.8.0-71-generic #71-Ubuntu SMP PREEMPT_DYNAMIC Tue Jul 22 16:52:38 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
alishba@ubuntu:~/analysis/workspace$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
D=ubuntu
D_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

```
alishba@ubuntu: ~/analysis/workspace

processor      : 1
vendor_id     : GenuineIntel
cpu family    : 6
model         : 142
model name    : Intel(R) Core(TM) i5-7300U CPU @ 2.60GHz
stepping      : 9
microcode     : 0xf6
cpu MHz       : 2712.002
cache size    : 3072 KB
physical id   : 2
siblings      : 1
core id       : 0
cpu cores     : 1
apicid        : 2
initial apicid : 2
fpu           : yes
fpu_exception : yes
cpuid level   : 22
wp            : yes
flags         : fpu vme de pse tsc msr pae mca cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc arch_p
rsmopl nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pt1 ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflushopt xsaveopt xsavec xge
tbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs          : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit srbds mmio_stale_data retbleed gds bhi
bogomips      : 5424.00
clflush size   : 64
cache alignment : 64
address sizes  : 45 bits physical, 48 bits virtual
power management:

alishba@ubuntu:~/analysis/workspace$ free -h
               total        used        free      shared  buff/cache   available
Mem:            1.9Gi         471Mi         875Mi         612Ki         756Mi         1.4Gi
Swap:           2.0Gi          68Mi         1.9Gi

alishba@ubuntu:~/analysis/workspace$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            192M  1.4M  191M   1% /run
/dev/sda2        30G   7.0G   21G  26% /
tmpfs            960M   0 960M   0% /dev/shm
tmpfs            5.0M   0  5.0M   0% /run/lock
tmpfs            192M  32K  192M   1% /run/user/1000
alishba@ubuntu:~/analysis/workspace$
```

```
alishba@ubuntu: ~/analysis/workspace

rsmopl nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pt1 ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflushopt xsaveopt xsavec xge
tbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs          : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit srbds mmio_stale_data retbleed gds bhi
bogomips      : 5424.00
clflush size   : 64
cache alignment : 64
address sizes  : 45 bits physical, 48 bits virtual
power management:

alishba@ubuntu:~/analysis/workspace$ free -h
               total        used        free      shared  buff/cache   available
Mem:            1.9Gi         471Mi         875Mi         612Ki         756Mi         1.4Gi
Swap:           2.0Gi          68Mi         1.9Gi

alishba@ubuntu:~/analysis/workspace$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            192M  1.4M  191M   1% /run
/dev/sda2        30G   7.0G   21G  26% /
tmpfs            960M   0 960M   0% /dev/shm
tmpfs            5.0M   0  5.0M   0% /run/lock
tmpfs            192M  32K  192M   1% /run/user/1000
alishba@ubuntu:~/analysis/workspace$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.6 22716 13184 ?        Ss   06:41   0:18 /usr/lib/systemd/systemd --system --deserialize=63
root         2  0.0  0.0   0     0 ?        S    06:41   0:00 [kthreadd]
root         3  0.0  0.0   0     0 ?        S    06:41   0:00 [pool_workqueue_release]
root         4  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-rcu_g]
root         5  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-rcu_p]
root         6  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-slab]
root         7  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-netns]
root        11  0.0  0.0   0     0 ?        I    06:41   0:03 [kworker/u256:0-ext4-rsv-conversion]
root        12  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-mm_pe]
root        13  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_kthread]
root        14  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_rude_kthread]
root        15  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_trace_kthread]
root        16  0.0  0.0   0     0 ?        S    06:41   0:01 [ksoftirqd/0]
root        17  0.0  0.0   0     0 ?        I    06:41   0:04 [rcu_preempt]
root        18  0.0  0.0   0     0 ?        S    06:41   0:00 [migration/0]
root        19  0.0  0.0   0     0 ?        S    06:41   0:00 [idle_inject/0]
root        20  0.0  0.0   0     0 ?        S    06:41   0:00 [cpuhp/0]
root        21  0.0  0.0   0     0 ?        S    06:41   0:00 [cpuhp/1]
root        22  0.0  0.0   0     0 ?        S    06:41   0:00 [idle_inject/1]
alishba@ubuntu:~/analysis/workspace$
```

Question 5

```
alishba@ubuntu: ~/analysis/workspace
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit srbds mmio_stale_data retbleed gds bhi
bogomips : 5424.00
clflush size : 64
cache_alignment : 64
address sizes : 45 bits physical, 48 bits virtual
power management:

alishba@ubuntu:~/analysis/workspace$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi        471Mi        875Mi        612Ki        756Mi        1.4Gi
Swap:         2.0Gi         68Mi         1.9Gi

alishba@ubuntu:~/analysis/workspace$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            192M  1.4M  191M   1% /run
/dev/sda2        30G   7.0G   21G  26% /
tmpfs            960M   0 960M   0% /dev/shm
tmpfs            5.0M   0  5.0M   0% /run/lock
tmpfs            192M  32K  192M   1% /run/user/1000

alishba@ubuntu:~/analysis/workspace$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.6 22716 13184 ?        Ss   06:41   0:18 /usr/lib/systemd/systemd --system --deserialize=63
root         2  0.0  0.0   0     0 ?        S    06:41   0:00 [kthreadd]
root         3  0.0  0.0   0     0 ?        S    06:41   0:00 [pool_workqueue_release]
root         4  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-rcu_g]
root         5  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-rcu_p]
root         6  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-slub_]
root         7  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-netns]
root        11  0.0  0.0   0     0 ?        I    06:41   0:03 [kworker/u256:0-ext4-rsv-conversion]
root        12  0.0  0.0   0     0 ?        I<   06:41   0:00 [kworker/R-mm_pe]
root        13  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_kthread]
root        14  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_rude_kthread]
root        15  0.0  0.0   0     0 ?        I    06:41   0:00 [rcu_tasks_trace_kthread]
root        16  0.0  0.0   0     0 ?        S    06:41   0:01 [ksoftirqd/0]
root        17  0.0  0.0   0     0 ?        I    06:41   0:04 [rcu_preempt]
root        18  0.0  0.0   0     0 ?        S    06:41   0:00 [migration/0]
root        19  0.0  0.0   0     0 ?        S    06:41   0:00 [idle_inject/0]
root        20  0.0  0.0   0     0 ?        S    06:41   0:00 [cpuhp/0]
root        21  0.0  0.0   0     0 ?        S    06:41   0:00 [cpuhp/1]
root        22  0.0  0.0   0     0 ?        S    06:41   0:00 [idle_inject/1]

alishba@ubuntu:~/analysis/workspace$ sudo adduser lab4user
[sudo] password for alishba:
fatal: The user `lab4user' already exists.
alishba@ubuntu:~/analysis/workspace$
```

```
fatal: The user `lab4user' already exists.
alishba@ubuntu:~/analysis/workspace$ getent passwd lab4user
lab4user:x:1001:1001:,,,:/home/lab4user:/bin/bash
alishba@ubuntu:~/analysis/workspace$
```

```
lab4user:x:1001:1001:,,,:/home/lab4user:/bin/bash
alishba@ubuntu:~/analysis/workspace$ su - lab4user
oami
pwd
Password:
lab4user@ubuntu:~$
```

```
lab4user@ubuntu:~$ exit
logout
alishba@ubuntu:~/analysis/workspace$
```

```
alishba@ubuntu:~/analysis/workspace$ sudo deluser --remove-home lab4user
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `lab4user' ...
alishba@ubuntu:~/analysis/workspace$
```